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UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
UTM Johor Bahru

SECP2613 – SYSTEM ANALYSIS AND DESIGN

SEMESTER II 2023/2024

KADA MANAGEMENT INFORMATION SYSTEM

ANALYSIS AND PLANNING

Phases:

Project Proposal and Planning &

Information System Gathering and Requirement

URL for Github repository : [Github Repository TechMedico](#)

URL for Github project : [Github Project TechMedico](#)

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1.0 Introduction

Due to the global covid-19 pandemic, there has been a shift from manual to online systems in a variety of industries, stressing the need for more convenient, safe, and efficient processes. This trend is notably noticeable in the registration system. Online registration not only ensures safety, speed, and convenience by avoiding personal interaction, but it also improves database security because the database is maintained on a highly secure server. In the contrary, information in paper form is vulnerable to multiple parties. Furthermore, online registration allows for real-time statistics updates. Organizers can access vital information, such as the number of members in the cooperative, at any time because the database is automatically updated.

One of KADA's departments is cooperatives, which are autonomous human organizations whose members join together to pursue common interests in the economic, social, and cultural spheres. It operates on numerous principles, one of which being democratic control by its members. Cooperatives are democratic organizations run by members who actively participate in policymaking and decision-making. Elected representatives are accountable to all members. The Basic Cooperative's members have equal voting rights and cooperatives at other levels are democratically constituted.

To address several problems with KADA cooperative's present manual registration process, we propose a system that **moves new member registration to an online mode**. Furthermore, the new system will **increase membership capacity**, allowing more people to benefit from the services. In addition, we will introduce an **automated annual report generation tool** to help management understand the cooperative's present situation.

2.0 Information gathering process

In our information collecting process, we used three key ways to acquire insights into the existing Lembaga Kemajuan Pertanian Kemubu (KADA) system. The methods are **interview** with KADA by having meeting, **observation** through website and **questionnaire** through Whatsapp.

2.1 Method used

First, we **joined an industrial sharing program** on 14 May 2024 through Webex platform where the KADA shared the detailed information on various aspects of their system. This information covered the topics such as the member registration process, loan application procedure, the loans available and the calculation of the loan. Additionally, we apply the **observation method by access the KADA website** to get more details of the system. The website provides addition information which allow us to gain a comprehensive understanding of the current processes and procedures of the system. We also chose an unobtrusive approach of information collecting which is **questionnaire**, which aids in the collection of critical information on human and system requirements. For the questionnaire preparation, our group collaborated with other groups to prepare various questions, including open-ended and closed questions. By asking open-ended questions, we may go deeper into the user's requirements and find other potential issues with the current system. Then, to acquire more precise information, we conduct a questionnaire with management using closed-ended questions.

2.2 Summary from method used

We have concluded the interview with KADA management on May 14, 2024, with a summary. The goal of this sharing program is to acquire information about the background and concept of KADA cooperative. We discussed the cooperative's functioning, the types of loans they supplied, how to apply for a loan, how the loan gets authorized, what the repayment rate is, and so on. As a result of this meeting, we decided to present **an online system** for KADA cooperative that **focuses on two major modules which are member registration and loans**. In addition, we conducted an observation on the KADA cooperative's current website. The aim of this observation is to examine the workflow efficiency and identify any problems on the website. We discover that the **website's navigation is challenging** and the user struggles to understand how to use the system. Aside from this, the **system lacks clarity**. There are no clear instructions on how to apply for benefits; they merely supply a form and do not indicate any subsequent actions. We also noticed **a delayed load time** during our observation. For the questionnaire, our group prepared two questions for the surveyor. This questionnaire session allowed us to gain a better understanding of the client's system requirements. The client also provided some further clarifications on our idea.



Figure 2.2.1 Current website of KADA cooperative

Assalamualaikum kpd pihak koperasi KADA. Group Saya (TechMedico)ingin menanyakan beberapa soalan

1. "Apakah kriteria khusus yang diambil kira oleh pihak koperasi dalam proses kelulusan pinjaman, dan berapa lama biasanya masa yang diambil dari permohonan sehingga kelulusan pinjaman diberikan?"
2. "Bolehkah pihak koperasi memberikan rekod sejarah kadar dividen yang telah dibayar kepada ahli dalam lima tahun terakhir?"

15:38

Figure 2.2.2 Example of the questions from our group

3.0 Background Study



Lembaga Kemajuan Pertanian Kemubu(KADA) is an agency under the Ministry of Agriculture and Food Security, established on March 30, 1972, under Section 4 (1) of Act 69, the Kemubu Agricultural Development Authority Act 1972, Lembaga Kemajuan Pertanian Kemubu (KADA) is an agency under the Ministry of Agriculture and Food Security. On March 2, 1973, YAB Tun Hj Abdul Razak Bin Hussein, the second prime minister of Malaysia, officially created it. KADA is a leading organization dedicated to improving the economic status of farmers and increasing the country's food production, especially in rice and paddy to fulfill domestic needs. To bring farmers' per capita earnings into line with national statistics, it seeks to help agricultural families. Additionally, KADA works to enhance agriculture in the Kemubu region and builds and strengthens farmer institutions.

KADA offers a wide range of services,we going to focus on cooperatives section that KADA held to open an investment space for the members registered under it.KADA offers its members that registered under the cooperative a variety of financial benefits,including investment such as “Modal Syer” and “Modal Yuran” ,loans to support personal and business needs.The are five main loan plan for member registered under it which are “Pinjaman Al Bai”, “Pinjaman Al Innah”, “Pinjaman skim khas”, “Pinjaman road tax and insurance” and “Pinjaman Al qardhul Hasan”In order to help members safely plan for the future, KADA also provides a savings plan created especially for retirement funds. These all services are designed to support members' financial development and stability, promoting a community of financially independent people.

Lembaga Kemajuan Pertanian Kemubu (KADA) cooperative(koperasi) will include service like investment, saving plan for retirement fund and a loan facility. Employees can apply for membership in this cooperative by paying a registration fee of RM 55 and investing RM 300 in unit shares. So, with a total of RM 355 to become a member. Members can acquire shares for RM300, which may be paid in lump sum or monthly salary deductions, with RM50 deducted over six months and an additional RM5 as a contribution to the member welfare fund. As required by the *Malaysian Cooperative Commission*, members must pay their share, up to a maximum of RM10,000, in order to qualify for loans. For membership fees, there is a minimum payment of RM35. KADA also differentiates between share capital, which is required for loan eligibility, and savings, which serve different purposes.

The system will not be able to allow the public to join its cooperative membership through the system. Membership is limited to those who register and fulfil the requirements established by KADA. Furthermore, KADA does not provide membership cards to members who have successfully registered as members. Serving registered members with perks and services designed only for them continues to be the major priority.

There is also many benefits to be involved in KADA cooperative offers the benefits which include a savings plan for retirement, with an annual dividend of approximately 4.2% on the shares. Additionally, members can apply for loans up to RM 20,000. Besides, member welfare fund supports member welfare activities including Hajj and Umrah payment, death benefits for members and their spouses, and other member welfare programmed. To help members after retirement and ensure their financial stability and future demands are met, fixed savings plans are also offered.

4.0 Problem Statement

The cooperative (koperasi) run by Lembaga Kemajuan Pertanian Kemubu (KADA) offers significant benefits to its employees, such as loans and a retirement fund. However, a cumbersome manual registration procedure and a restrictive membership limit are the cooperative's two main problems, which restrict its efficiency and accessibility.

At this point, **employees must fill out a manual form to register as a member of the KADA cooperative**. This outdated method is hard to use and results in a lot of mistakes and inefficiency. Employees must complete paper forms, which can be cumbersome and error prone. Employees are discouraged from enrolling by this cumbersome process as they may think it is too complex or inconvenient. Additionally, the registration form can only be found in Malay, which presents a major problem in increasing the number of employees registered under the cooperative. This is because they might be some employees do not know the Malay language well. The cooperative's reach is restricted by this linguistic barrier, which also decreases the benefits of KADA's diversity and accessibility.

Apart from the registration process, which is cumbersome, there is **a 500 members cap on the cooperative** now. This restriction significantly limits the number of employees who may register as a member of the cooperatives. Many potential employees are unable to join and benefit from the cooperative's financial advantages since the workforce exceeds the membership the limit. Due to this restriction, KADA's employees is unable to take advantage of important loan activities and retirement savings plans that might improve their well-being and financial stability. The membership cap impacts not just the options available to individuals but also the cooperative's ability to develop and have an influence.

The third main problem that KADA cooperative must overcome is the **lack of month-end and year-end financial reports**, which include profit and loss statements. Because members are unable to monitor the cooperative's financial status or the management of their contributions, the lack of financial transparency affects member

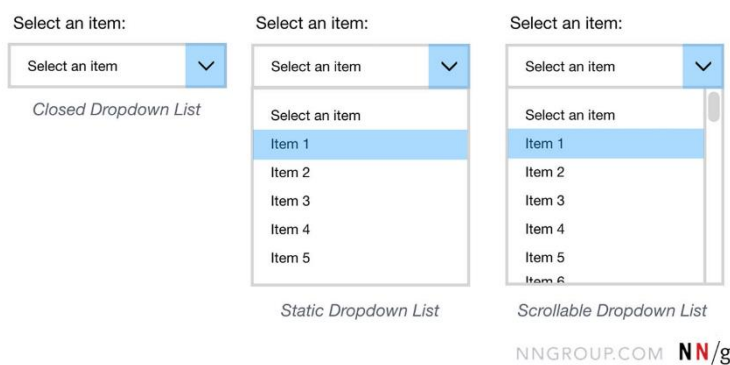
confidence and involvement. Improving the cooperative's efficiency, dependability, and attraction for current and new members will involve solving this issue.

These issues have wide-ranging consequences inefficiencies of the manual registration procedure, number of members restriction and system that doesn't has a summary report inhibit employees from joining the cooperative. As a result, the cooperative's membership decreases and clients' general satisfaction with its services declines. Due to the difficulty and barrier to registration, employees who may profit from the cooperative's financial goods may be left out. Furthermore, this problem is made worse by the membership cap, which keeps many motivated employees from taking part, missing out on opportunities for money, and decreasing productivity. The cooperative's limited capacity to provide services to a larger portion of KADA's workforce reduces its chances for development and lowers its efficacy as a financial assistance system for employees.

5.0 Proposed Solutions

The present manual registration method has issues that the **e-registration system** was created to solve this problem. Every operation and information in this system is computerized. It's essential to take into consideration the feasibility of the proposed system which are operational, technological, and financial aspects to solve the difficulties the KADA cooperative is facing and provide an achievable solution.

Operation Feasibility: Our suggestion is to change from a **manual registration procedure to an online** automated method to make it simpler and quicker. The current system is prone to errors and inefficient, as it requires physical forms. We suggest that the system can mainly using drop down box to provide selection for the register members so that users do not need to enter all the information on their own. This can reduce the chance of typos and make the Data Mining process. We also suggest that the registration form should also be provided in **Chinese, English, and Malay**. To enable more employees to join, we suggest **increasing the membership limit from 500 to a larger figure**. This requires altering methods to efficiently handle additional members by establishing a offer of multilingual guidance in the registration. Other than that, is enhanced the financial reporting to evaluate the cooperative's financial situation, particularly its profit or loss. Transparency and accountability within the cooperative will be promoted by **making financial reports available to members via the web system**.



Example of drop-down box

Technical Feasibility: Creating a secure online platform is crucial. This platform needs to support financial information and member privacy will be ensured by using **secure authentication and robust encryption to ensure data security and privacy.**

In **Software**-wise, Microsoft Word will be used for financial reporting, making it possible to create and handle thorough financial documentation. The user interface will be designed and prototyped using Figma and Lucidchart to do flowchart.



For hardware, in order to create and execute the system, the minimum specifications for a laptop or computer include having at least Windows 7 as the operating system. Moreover, the system requires at least Intel i3 processor to make sure smooth execution of performance. Additionally, a minimum of 4GB of RAM is necessary to handle the system operation efficiently. For storage, the recommended specification should be at least 256GB of hard drive storage for sufficient data capacity.

These hardware and software elements working together will provide a reliable and secure system that satisfies the requirements of the cooperative.





6.0 Objectives

The primary objective of this project is to modernize and digitize Koperasi KADA's operations by following the Software Development Life Cycle (SDLC) methodology.

In the **Information Gathering** phase, the goal is to collect comprehensive information about current processes, stakeholder needs, and system requirements. This involves conducting interviews, surveys, and reviewing existing documentation to identify pain points and areas for improvement.

Following this, in the **Conduct Analysis Based on the Information Gathered** phase, the collected data will be analysed to understand the requirements and scope. This includes performing a gap analysis, defining functional and non-functional requirements, and prioritizing these requirements based on business impact.

Next, in the **Design Phase**, the objective is to create detailed designs for the digital systems. This encompasses developing system architecture, user interfaces, and data models. The design documentation will then be reviewed with stakeholders to ensure it aligns with their requirements.

The **Implementation Phase** involves developing and deploying the digital systems according to the design specifications. Integration with existing infrastructure, testing, and debugging will be carried out to ensure seamless operation.

To ensure quality, the **Testing and Quality Assurance** phase will ensure that the digital systems are functional, reliable, and meet all specified requirements. This will involve conducting system testing, integration testing, and user acceptance testing (UAT) to identify and fix defects.

Once the systems are ready, the **Deployment and Launch** phase will deploy the developed systems into the live environment. This phase includes providing training and support to end-users and monitoring the systems post-deployment for any issues.

Finally, in the **Maintenance and Evaluation** phase, the systems will be maintained and their performance evaluated over time. Updates and improvements will be implemented as needed based on user feedback and performance metrics.

7.0 Scope of the Project

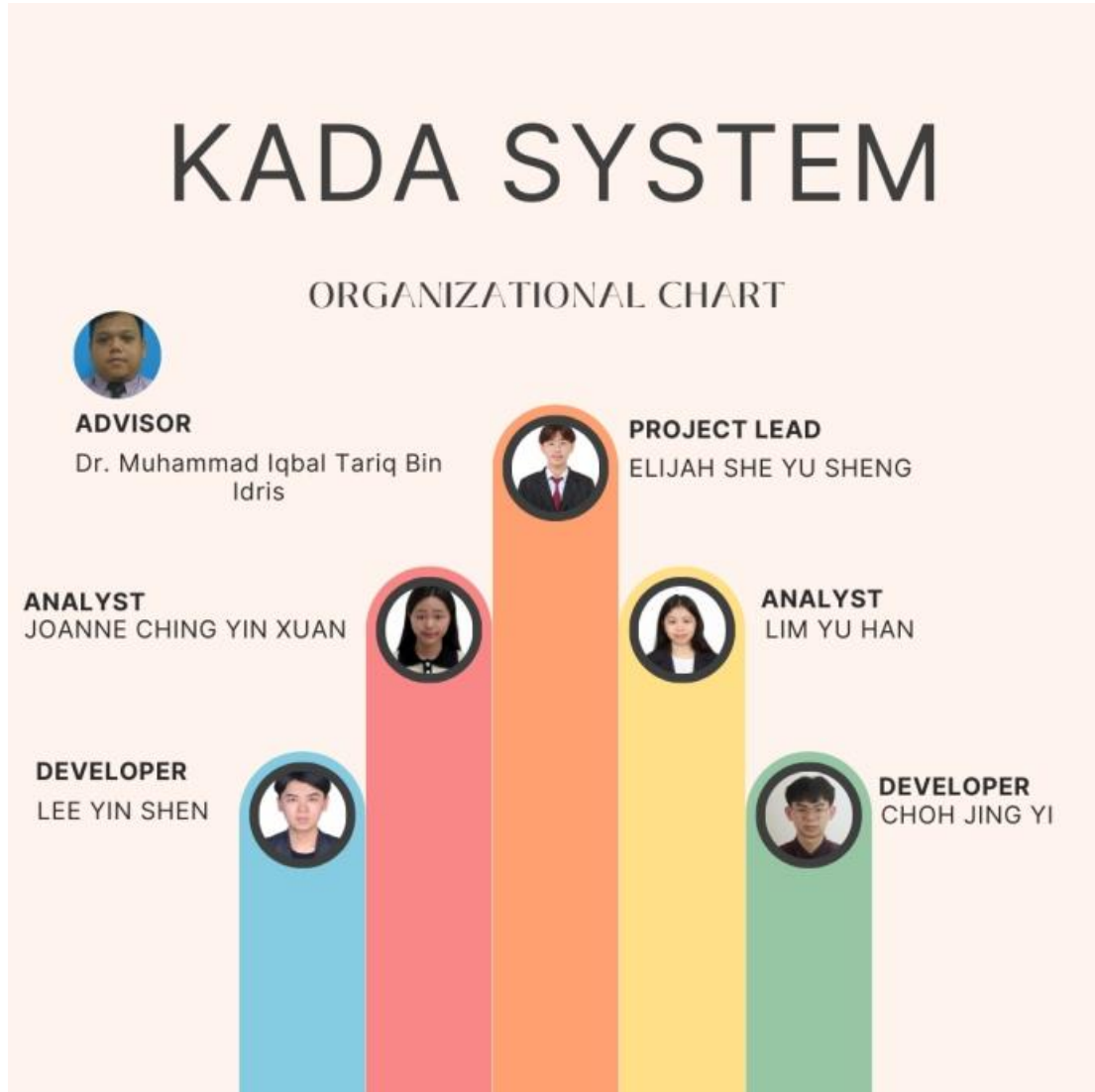
The digital transformation project for Koperasi KADA will have several limitations to ensure it remains manageable and focused. **In terms of technology**, the project will primarily concentrate on developing web-based solutions. While these systems will significantly enhance accessibility and streamline various processes, they will be limited to web applications only, excluding the development of mobile applications. Users will need access to a web browser to utilize the new digital systems, as mobile application development is not included within the project's scope. Additionally, all interactions and processes facilitated by the new systems will be conducted online, meaning any offline or manual processing requirements are outside the project's scope. Members and administrators must have internet access to participate in the cooperative's digital transformation.

In terms of actors, the new digital systems will be designed for use by specific roles within Koperasi KADA, including the Board of Directors, administrative staff, committee members, and new cooperative members. These systems will cater to the needs of these user groups, ensuring they can efficiently perform their tasks within the cooperative's operations. The project will focus on optimizing at least five key processes: registering new members, simplifying the loan application and approval process, enhancing reporting capabilities, establishing effective communication channels, and digitizing board meetings to improve efficiency and record-keeping.

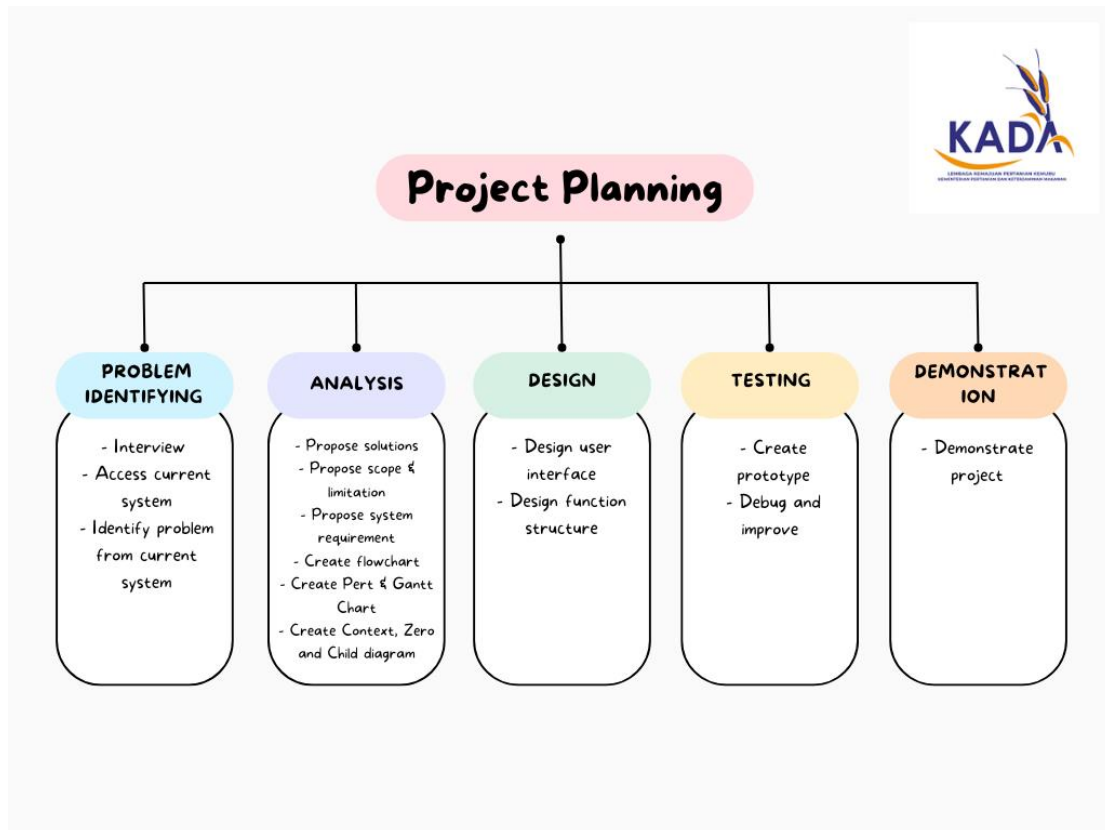
These limitations are set to ensure that the project remains manageable and focused, providing clear boundaries within which the digital transformation efforts will be implemented. By concentrating on web-based solutions and specific user roles, Koperasi KADA can achieve meaningful improvements in their operations while laying a foundation for potential future expansions, such as mobile applications or offline capabilities.

8.0 Project Planning

8.1 Human Resource



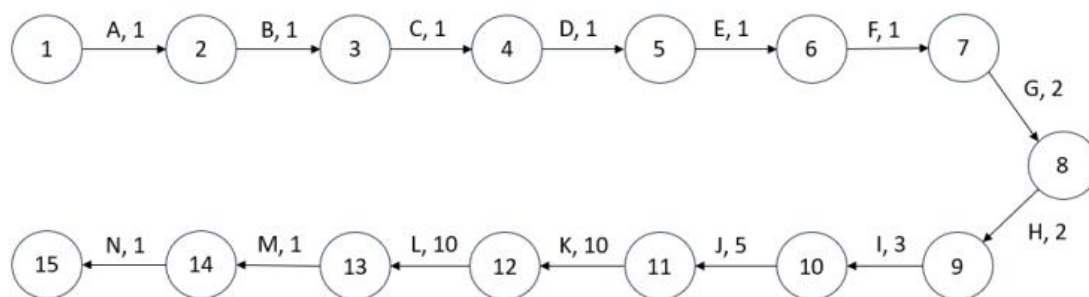
8.2 Work Breakdown Structure (WBS)



8.3 PERT Chart (based on WBS)

Pert Diagram

Activity		Duration (days)
A	Interview	1
B	Assess Current System	1
C	Identify Problems from Current System	1
D	Propose solutions	1
E	Propose scope & limitation	1
F	Propose system requirement	1
G	Create flow chart	2
H	Create Pert & Gantt chart	2
I	Create Context, Zero and Child Diagram	3
J	Design user interface	5
K	Design function structure	10
L	Create prototype	10
M	Debug and improve system	1
N	Demonstrate project	1



Path: A-B-C-D-E-F-G-H-I-J-K-L-M-N

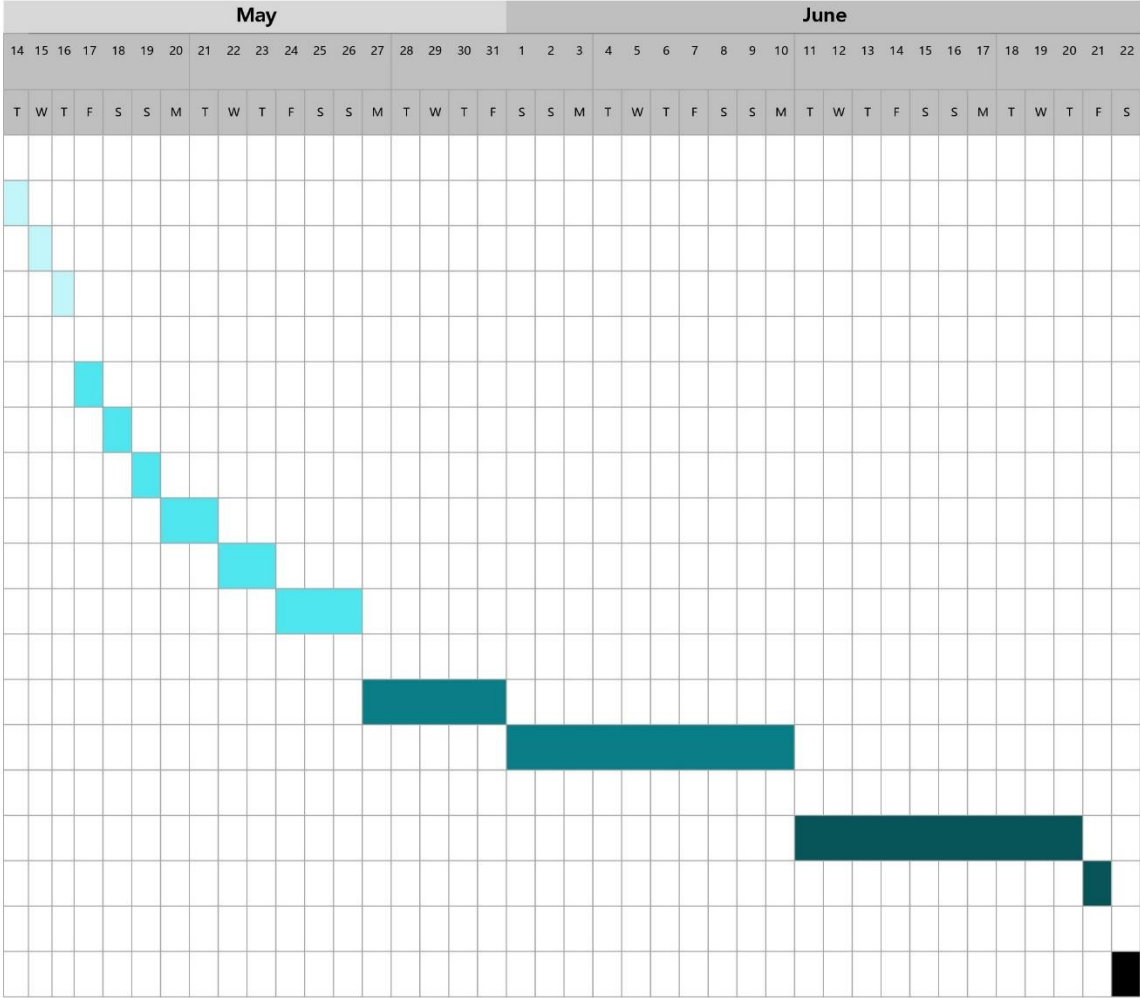
Length: 1+1+1+1+1+1+2+2+3+5+10+10+1+1

Total duration: 40 days

8.4 Gantt Chart

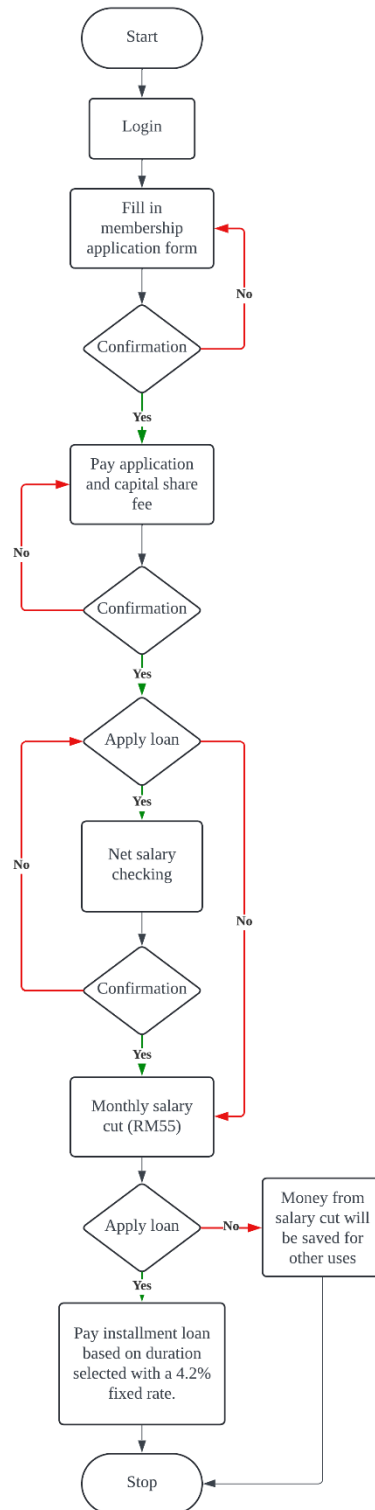
Project Title: KADA MIS
Project Start Time: 14/5/2024

Process	Start	Duration (day)
Problem Identification	14/5/2024	
Interview	14/5/2024	1
Access current system	15/5/2024	1
Identify problem from current system	16/5/2024	1
System Analysis	17/5/2024	
Propose solution	17/5/2024	1
Propose scope & limitation	18/5/2024	1
Propose system requirement	19/5/2024	1
Create flowchart	20/5/2024	2
Create Pert & Gantt chart	22/5/2024	2
Create Context, Zero and Child diagram	24/5/2024	3
System Design	27/5/2024	
Design User Interface	27/5/2024	5
Design function stucture	1/6/2024	10
Testing	11/6/2024	
Create prototype	11/6/2024	10
Debug and improve	21/6/2024	1
Project Demonstration	22/6/2024	
Demonstrate project	22/6/2024	1



9.0 Requirement Analysis (based on AS-IS analysis)

9.1 Current business process (scenarios, workflow)



Scenario 1: Mr. Haikal want to start an agriculture business but lack of capital

Mr. Haikal wants to start an agriculture business but lack of capital. Therefore, Mr. Haikal decides to apply RM 10000 loan from the Lembaga Kemajuan Pertanian Kemubu (KADA). In order to apply for the loan, Mr. Haikal must first become a member of the KADA. First, Mr. Haikal must access the KADA website and login the system using username and password. Then, Mr. Haikal must download the membership application form and fill in the form with his personal details like his name, phone number, ID card number, marriage status, home address, job position, net salary and etc. After fill in the form, Mr. Haikal need to submit it and wait for confirmation. The KADA will approve the registration if Mr. Haikal is qualified in a month. After getting approvment, Mr. Haikal will receive a confirmation call and he need to pay RM50 for member fee, RM5 for member welfare fund and RM300 for unit shares, with a total RM355. The RM50 should be pay in a maximum of 6 months while the RM300 can be paid in one lump sum or in installments. After becoming a member, Mr. Haikal need to download the loan application form from KADA website. Then he can choose a loan type to apply the RM10000 loan and choose the duration for installments payment. For example, he choose AL-INAH loan, so he need to pay the loan with a fixed rate 4.2%. In this case, if Mr. Haikal choose 1 year duration for installment payment, he need to pay RM868.33 per month. Therefore, he need to pay a total of RM923.33 per month which include the RM55 for member fee and RM868.33 for installments.

Scenario 2: Mrs. Halimah is facing financial problem for her children's school expenses

Mrs. Halimah is a single mother who are having 2 children who are currently about to enter primary school. She is facing financial problem for her children's school expenses. Therefore, she decides to apply loan for about RM2000 from KADA. So, she firstly needs to become a member before applying the loan. Firstly, she needs to obtain the member registration form from KADA website, print and fill the form manually with her personal information such as name, ID card number, races, job position and etc. Then, she must submit the form to office and wait for confirmation in a month. After being approved, she needs to download the loan application form again from the KADA website, fill and submit manually to office. In this form, she can choose loan type, for example she chooses AL-BAI loan. For this loan type, there is a 4.2% fixed installment rate, and in this case let say she choose duration of 3 years. Therefore, she needs to pay RM62.56 for installment, and RM55 for member fee, so total is RM117.56 monthly.

9.2 Functional Requirement (input, process and output)

The functional requirements for the KADA system involve distinct process for **both user and admin**. For user, the input include **submit a membership application form** which filled with their personal information, and a **loan application form** which filled with the loan type and duration for installment payments. Then, the system will processes that input by verifying the completeness and validity of the user's detail, and calculating the associated membership fees. Upon approval, the system process the loan application and calculate the total loan payment based on the selected loan type and duration. The output for the user include receiving a confirmation of membership approval and a payment notice of membership fee and a loan payment schedule notice which show the total monthly payment amount.

For admin, the input include **accessing the submitted membership and loan application forms by the user**. The admin will process these inputs by reviewing and verifying the membership application form to make sure all user information is valid, then approve or reject the membership application based on the qualification criteria. Furthermore, the admin will review and verify the loan application forms, then approve and reject them based on the qualification criteria and user detail, and also calculate the loan payment and interest based on the chosen loan type and duration. The output for admin is giving notifications to user about the status of their membership and loan application, the confirmation of payment receipt and the amount of installment payment.

9.3 Non-functional Requirement (performance and control)

The application for the loan must be easy to use and understand for the employees and admin. It requires users to log in with a password to ensure they are KADA employees and to protect their account details. The system also needs to ensure that the login process does not take a long time. Instructions on the website need to be clear so that users do not confuse the loan application with cooperative member registration. Additionally, the system should automatically calculate the payment amount, and users should have the flexibility to choose their preferred payment methods.

9.4 Logical DFD AS-IS system (Context Diagram, Diagram 0, Child)

Context diagram of KADA MANGEMENT INFORMATION SYSTEM

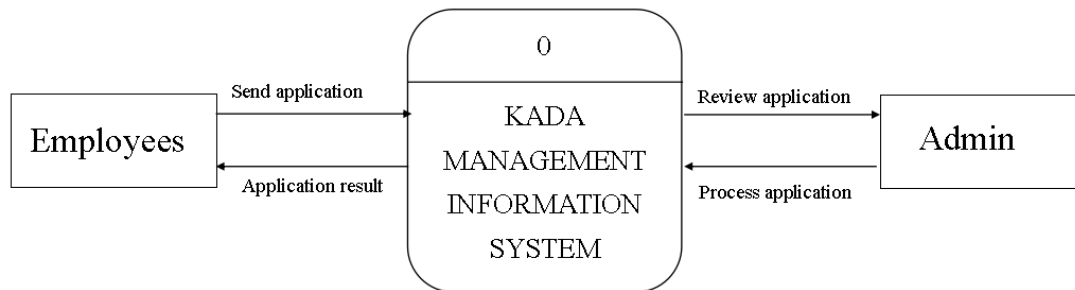
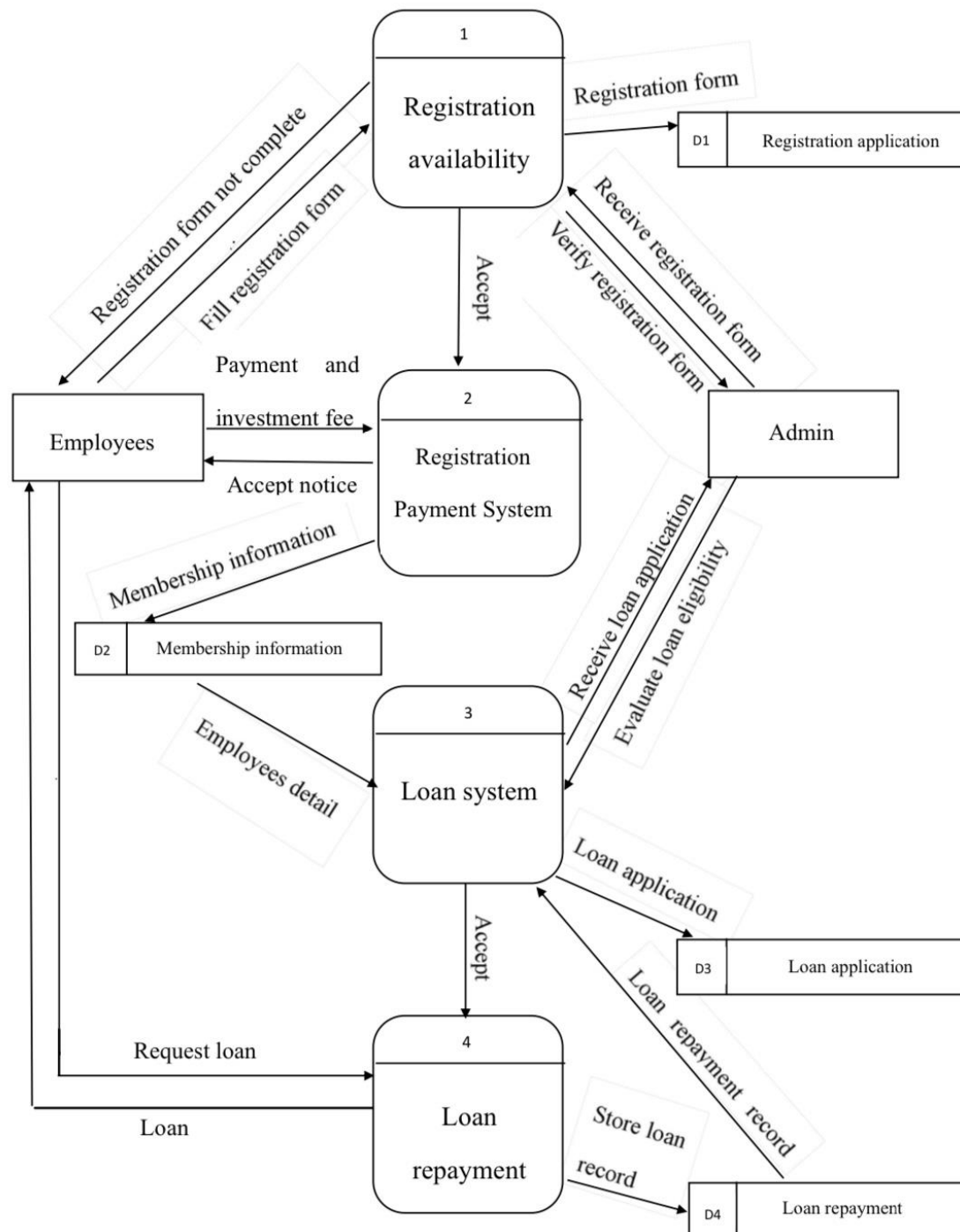
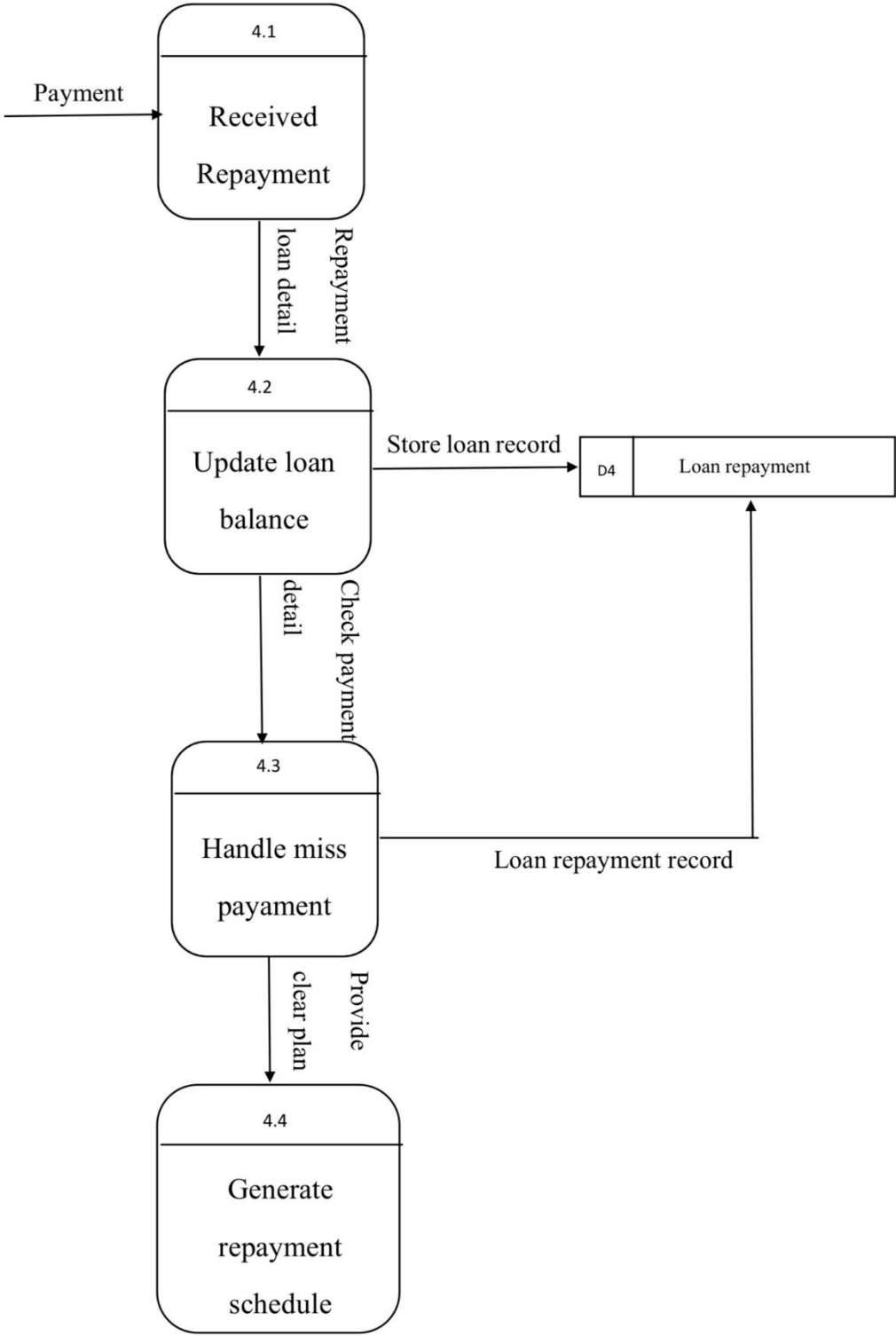


Diagram 0 of KADA MANGEMENT INFORMATION SYSTEM



Child diagram



10.0 Benefit and Overall Summary of Proposed System

KADA.gov.my is a system that we proposed to address the inefficiencies and limitations of the current manual registration process. Firstly, the system can enhance membership accessibility to broaden its reach and support a larger member base. The cooperative can broaden its potential membership base and reach more people by assessing the viability of introducing membership to the public. Additionally, digitizing Board of Directors meetings can significantly improve efficiency. By using a digital system for these meetings, it will be easier to keep track of minutes, make the sessions more transparent and efficient for all board members, and streamline record-keeping.

Furthermore, the system's suggestion to streamline the loan approval and membership processes. The current manual processes should be evaluated and digitised to achieve faster processing times, lower member wait times, and increasing overall satisfaction. Effective communication lines between the cooperative and its members are crucial for keeping everyone up to date on their rights, benefits, and any policy changes. Finally, the members will find it easier to join and apply for loans at any time and from any location by adopting an online application system that will improve accessibility and speed the application process.

Overall, the new online application system will significantly improve member satisfaction by simplifying the registration and loan application processes. The Lembaga Kemajuan Pertanian Kemubu will be in a better position to assist its members and accomplish the organization's goal of improving the socioeconomic status of farmers in the Kemubu region by implementing these reforms. This modernization initiative will also guarantee that KADA's cooperative is safe, secure, and responsive to the demands of its members.

11.0 Repository Snapshot

TechMedico_Project_SAD_20232024Public

Watch1Fork3Star0

main1 Branch0 Tags

Go to file

Add fileCode

leeyinshen0818 Update README.md620ff4e · now208 Commits

1.0 Introduction	Create 1.0	1 hour ago
10.0 Benefit and Overall Summary of Proposed...	Update Benefit and Overall Summary of Proposed System.md	25 minutes ago
2.0 Information Gathering Process	Update 2.2	5 minutes ago
3.0 background	Update 3.0 Background report.md	22 minutes ago
4.0 Problem Statement	Create img	21 minutes ago
5.0 Proposed Solutions	Update 5.0 Proposed Solutions report.md	31 minutes ago
6.0 Objectives	Update objectives.md	17 minutes ago
7.0 Scope of the project	Update scope.md	14 minutes ago
8.0 Project Planning	Update 8.4 Gantt Chart.md	30 minutes ago
9.0 Requirement Analysis	Update 9.4 Logical DFD AS-IS system (Context Diagram, Dia...	26 minutes ago
img.file	Add files via upload	7 minutes ago
README.md	Update README.md	now

About

No description, website, or topics provided.

Readme

Activity

0 stars

1 watching

3 forks

Report repository

Releases

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Packages

No packages published

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